

Solutions to Problem Set 3

1. We make the change of variable $u = \ln x$, which is the same as $x = e^u$. Then $dx = e^u/du$ and as x varies from e to ∞ , u varies from 1 to ∞ . Therefore we have

$$\int_e^\infty \frac{dx}{(\ln x)^p} = \int_1^\infty u^{-p} e^u du.$$

For every real p , we have

$$u^{-p} e^u \rightarrow \infty \text{ as } u \rightarrow \infty.$$

The integral in question diverges for every real p . We can also see that the integral diverges by using the Fact at the bottom of page 29 from Lecture 3. It is clear that we have divergence for $p \leq 0$. For $p > 0$, we choose $a, c > 0$ such that

$$e^{-u} \leq \frac{c}{u^p}, \quad u \geq a.$$

This tells us that

$$u^{-p} e^u \geq \frac{1}{c}, \quad u \geq a,$$

and this implies that the integral diverges.

2. The integral in question converges. To see this, we make the change of variable $u = \ln x$, which is the same as $x = e^u$. Then $dx = e^u/du$ and as x varies from e to ∞ , u varies from 1 to ∞ . Therefore

$$\int_e^\infty \frac{dx}{(\ln x)^{\ln x}} = \int_1^\infty \frac{e^u}{u^u} du.$$

Notice that

$$\frac{e^u}{u^u} \leq e^{-u}, \quad u \geq e^2.$$

Since

$$\int_{e^2}^\infty e^{-u} du$$

converges, we deduce that

$$\int_1^\infty \frac{e^u}{u^u} du$$

by the comparison test.

3. (a) Since $\vec{x} \cdot \vec{y} = 0$, we have

$$|\vec{x} - \vec{y}|^2 = \vec{x} \cdot \vec{x} - 2\vec{x} \cdot \vec{y} + \vec{y} \cdot \vec{y} = \vec{x} \cdot \vec{x} + \vec{y} \cdot \vec{y}.$$

We also have

$$|\vec{x} + \vec{y}|^2 = \vec{x} \cdot \vec{x} + 2\vec{x} \cdot \vec{y} + \vec{y} \cdot \vec{y} = \vec{x} \cdot \vec{x} + \vec{y} \cdot \vec{y} = |\vec{x} - \vec{y}|^2.$$

Since $|\vec{x} - \vec{y}| \geq 0$ and $|\vec{x} + \vec{y}| \geq 0$, we conclude that

$$|\vec{x} - \vec{y}| = |\vec{x} + \vec{y}|.$$

(b) Since $|\vec{v} - \vec{w}|^2 = |\vec{w}|^2$, we find that

$$\vec{v} \cdot \vec{v} - 2\vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{w} = \vec{w} \cdot \vec{w}.$$

Using the fact that $\vec{v} \cdot \vec{v} = 1$, we conclude that

$$1 - 2\vec{v} \cdot \vec{w} = 0,$$

which yields $\vec{v} \cdot \vec{w} = \frac{1}{2}$.

4. Observe that

$$\vec{F}'(t) = e^t(\cos t - \sin t, \sin t + \cos t, 1)$$

and that

$$(\cos t - \sin t)^2 + (\sin t + \cos t)^2 = 2(\sin^2 t + \cos^2 t) = 2.$$

We conclude that

$$|\vec{F}'(t)| = e^t \sqrt{2+1} = \sqrt{3}e^t,$$

and consequently

$$\int_0^\pi |\vec{F}'(t)| dt = \sqrt{3} \int_0^\pi e^t dt = \sqrt{3}(e^\pi - 1).$$

5. (a) $\frac{\partial}{\partial x} \left(e^{-x^4} y^2 \ln(1 + y^2) \right) = -4x^3 e^{-x^4} y^2 \ln(1 + y^2)$

(b) $f_2(a, b, c) = \frac{a^2 + b^2 + c^2}{a + b + c} + 2b \ln(a + b + c)$

(c) Here it is best to change the order of differentiation and take the y derivatives first. We have

$$f_y(x, y) = 2ye^{-x^3}, \quad f_{yy}(x, y) = 2e^{-x^3}, \quad f_{yyy}(x, y) = 0.$$

We conclude that

$$f_{xxxxxyy}(x, y) = f_{yyyxxx}(x, y) = 0.$$

(d) $u_{x_3}(x_1, x_2, x_3) = \frac{3}{2} \sqrt{x_1 x_2^2 x_3}$

(e) $\frac{\partial x}{\partial r} = \cos \theta, \quad \frac{\partial x}{\partial \theta} = -r \sin \theta$

(f)

$$f_x(x, y) = 2x \cos(x^2 + 3y) + 2xy^3$$

$$f_{xx}(x, y) = -4x^2 \sin(x^2 + 3y) + 2 \cos(x^2 + 3y) + 2y^3$$

$$f_{xy}(x, y) = -6x \sin(x^2 + 3y) + 6xy^2$$

6. We begin by observing that

$$g_1(\vec{x}) = 3x_1^2x_2 + 2x_3^2, \quad g_2(\vec{x}) = x_1^3 + 2e^{x_2}, \quad g_3(\vec{x}) = 4x_1x_3.$$

(a) $\vec{\nabla}g(\vec{x}) = (3x_1^2x_2 + 2x_3^2, x_1^3 + 2e^{x_2}, 4x_1x_3)$

(b) We have $g(1, 0, 2) = 9$ and

$$\vec{\nabla}g(1, 0, 2) = (8, 3, 8),$$

so that the linear approximation of g about $(1, 0, 2)$ is given by

$$9 + (8, 3, 8) \cdot (\vec{x} - (1, 0, 2)) = 9 + 8(x_1 - 1) + 3x_2 + 8(x_3 - 2).$$

(c) The exact value is given by

$$g(1.01, .02, 1.97) = 8.900834794.$$

The linear approximation gives

$$g(1.01, .02, 1.97) \approx 8.9$$

7. Since the second-order partials are continuous, we know that the mixed second-order partials are equal for u and v . Therefore, we have

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial y \partial x} = 0,$$

and

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = -\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y \partial x} = 0.$$

8. We have

$$f_x(x, y) = 6x - 3y^2, \quad f_y(x, y) = -6xy + 3y^2 + 6y, \quad f_{xx}(x, y) = 6,$$

$$f_{xy}(x, y) = f_{yx}(x, y) = -6y, \quad f_{yy}(x, y) = -6x + 6y + 6.$$

Substituting in the numbers we find that $H_f(-1, 1)$ is given by

$$H_f(-1, 1) = \begin{pmatrix} 6 & -6 \\ -6 & 18 \end{pmatrix}$$

9. Let us fix $\vec{x} \in \mathbb{R}^n$ and define $\vec{u} : \mathbb{R} \rightarrow \mathbb{R}^n$ by

$$\vec{u}(t) = t\vec{x} \text{ for all } t \in \mathbb{R}.$$

Observe that

$$\vec{u}'(t) = \vec{x} \text{ for all } t \in \mathbb{R}.$$

By the chain rule, we deduce that

$$\frac{d}{dt}f(t\vec{x}) = \frac{d}{dt}f(\vec{u}(t)) = \vec{\nabla}f(\vec{u}(t)) \cdot \vec{u}'(t) = \vec{\nabla}f(t\vec{x}) \cdot \vec{x}.$$

Observe also that

$$\frac{d}{dt}(t^m f(\vec{x})) = mt^{m-1}f(\vec{x}) \text{ for all } t \in \mathbb{R}.$$

It follows that

$$\vec{\nabla}f(t\vec{x}) \cdot \vec{x} = mt^{m-1}f(\vec{x}) \text{ for all } t \in \mathbb{R}.$$

Putting $t = 1$ in the expression above gives

$$\vec{\nabla}f(\vec{x}) \cdot \vec{x} = mf(\vec{x}).$$

10. We use the chain rule and the product rule.

$$\frac{\partial w}{\partial x} = f'(u) \frac{\partial u}{\partial x} = f'(u) \cdot 2x,$$

$$\frac{\partial^2 w}{\partial x^2} = f'(u) \cdot 2 + f''(u) \cdot 2x \cdot \frac{\partial u}{\partial x} = 2f'(u) + 4x^2 f''(u),$$

$$\frac{\partial w}{\partial y} = f'(u) \frac{\partial u}{\partial y} = f'(u) \cdot 2y,$$

$$\frac{\partial^2 w}{\partial y^2} = 2f'(u) + 4y^2 f''(u).$$

Adding the appropriate equations gives

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = 4f'(u) + 4(x^2 + y^2)f''(u) = 4f'(u) + 4uf''(u).$$